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Batch : B3(2025-2029)

ASSIGNMENT No: 14

TITLE: Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Problem Statement

Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory

Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

Code

```
import java.io.*;

public class ExceptionDemo{
    public static void main(String []a){
        try{
            int arr[]={10,20,30,40};
            try{

                try{

                    int result=arr[5];
                    System.out.println(result);
                }catch(ArrayIndexOutOfBoundsException e){

                    System.out.println("Nested Catch : Array Index Out of Bound:"+e.getMessage());
```

```
}

int num=10/0;
System.out.println(num);

}catch(ArithmeticException e){
System.out.println("Outer Catch : Arithmetic Exception:"+e.getMessage());
}

}catch(Exception e){
System.out.println("General Exception:"+e.getMessage());
}

System.out.println("program continues..");
}
}
```

Output

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

D:\DSY JAVA\E14>javac ExceptionDemo.java

D:\DSY JAVA\E14>java ExceptionDemo
Nested Catch : Array Index Out of Bound:5
Outer Catch : Arithmetic Exception:/ by zero
program continues..

D:\DSY JAVA\E14>_
```

The screenshot shows a Windows Command Prompt window with the title bar "C:\Windows\System32\cmd.exe". The window displays the output of a Java program. The first command is "javac ExceptionDemo.java", which compiles the program. The second command is "java ExceptionDemo", which runs the program. The output shows two catch blocks: "Nested Catch : Array Index Out of Bound:5" and "Outer Catch : Arithmetic Exception:/ by zero". The program then prints "program continues..". The window also shows the system tray at the bottom with the date "06-08-2026", time "12:07", and a notification icon with the number "4".

Exercise

1. Create an ATM program handling invalid withdrawal amount using try-catch

Code

```
import java.io.*;
import java.util.Scanner;

public class ATMDemo {
    public static void main(String[] args) {
        int balance = 5000;
        Scanner scanner = new Scanner(System.in);

        System.out.println("Available Balance: " + balance);
        System.out.print("Enter amount to withdraw: ");

        try {
            int amount = scanner.nextInt();

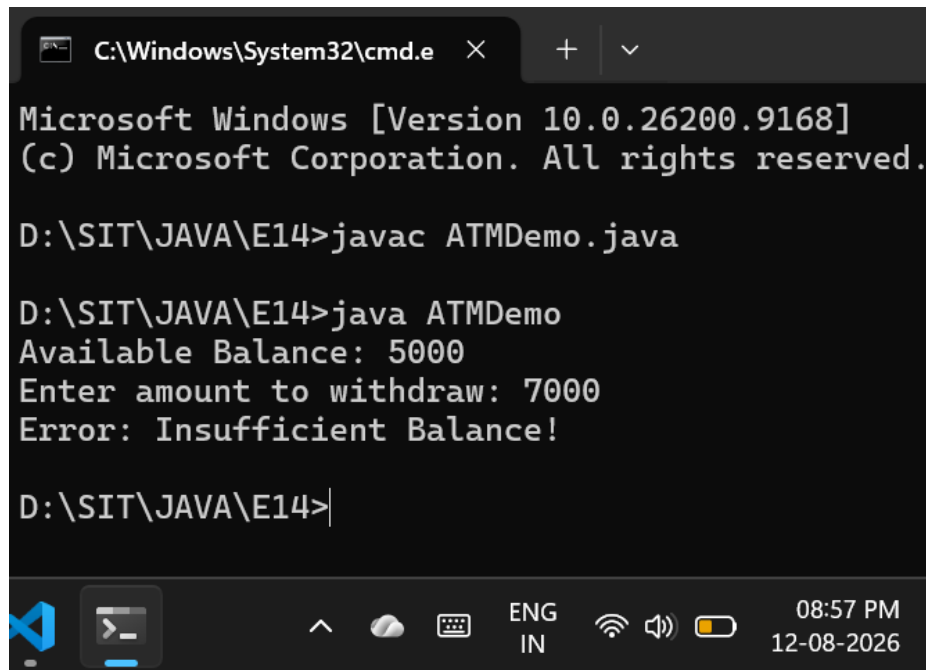
            if (amount > balance) {
                throw new IllegalArgumentException("Insufficient Balance!");
            } else if (amount <= 0) {
                throw new IllegalArgumentException("Amount must be greater than zero!");
            }

            balance = balance - amount;
            System.out.println("Please collect your cash.");
            System.out.println("Remaining Balance: " + balance);

        } catch (IllegalArgumentException e) {
            // Catches the invalid amount error
            System.out.println("Error: " + e.getMessage());
        } catch (Exception e) {
            System.out.println("Error: Invalid input! Please enter a number.");
        }

        scanner.close();
    }
}
```

Output



```
C:\Windows\System32\cmd.e

Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

D:\SIT\JAVA\E14>javac ATMDemo.java

D:\SIT\JAVA\E14>java ATMDemo
Available Balance: 5000
Enter amount to withdraw: 7000
Error: Insufficient Balance!

D:\SIT\JAVA\E14>
```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

Code

```
import java.io.*;

public class OnlineShopping {

    void processOrder(int quantity) {
        int pricePerItem = 500;

        if (quantity <= 0) {
            throw new IllegalArgumentException("Invalid Quantity! Quantity must be 1 or
more.");
        }

        int totalCost = quantity * pricePerItem;
        System.out.println("Success! You ordered " + quantity + " items. Total: " + totalCost);
    }

    public static void main(String[] args) {
        OnlineShopping s = new OnlineShopping();
    }
}
```

```
try{
    System.out.println("\nfirst Order");
    s.processOrder(2);
} catch (IllegalArgumentException e) {
    System.out.println("Error: " + e.getMessage());
}

System.out.println("\nSecond Order");
try{

    s.processOrder(0);
} catch (IllegalArgumentException e) {
    System.out.println("Error Caught: " + e.getMessage());
}

System.out.println("\nProgram ended.");
}
}
```

Output

```
D:\SIT\JAVA\E14>javac OnlineShopping.java
D:\SIT\JAVA\E14>java OnlineShopping

first Order
Success! You ordered 2 items. Total: 1000

Second Order
Error Caught: Invalid Quantity! Quantity must be 1 or more.

Program ended.

D:\SIT\JAVA\E14>
```

